INDUSTRY PRACTICES USING THE SDLC

Requirements Engineering

Understand the problem before building the solution.

DISCUSS FIRST

If a client asks for an app, what do you do first?

Think • Discuss • Share

Start with the problem, not the code

A feature request is a starting point—not a complete specification.

- Who needs the system?
- What are they trying to accomplish?
- What problem exists today?
- What would success look like?

DISCUSS FIRST

What are software requirements?

Think • Discuss • Share

Requirements define the system

- Needs and capabilities: what the system must do
- Constraints and quality expectations: how it must behave
- Stakeholders: whose needs matter
- Acceptance criteria: how we know it works

A client says: “Build me a restaurant app.”

What is still unknown?

01

The request

“I want an application for my restaurant.”

02

Engineering questions

Ordering or reservations? Who uses it?
Payments? Delivery? Cancellations?

DISCUSS FIRST

Why do requirements matter?

Think • Discuss • Share

Unclear requirements create real costs

- Building the wrong features
- Misunderstood behavior and expensive rework
- Missing user groups and exceptional cases
- Delays, frustrated customers, and avoidable risks

DISCUSS FIRST

What types of requirements exist?

Think • Discuss • Share

Four useful perspectives

FUNCTIONAL

What must it do?

NON-FUNCTIONAL

How well must it work?

BUSINESS

Why does the organization need it?

USER

What must the person achieve?

DISCUSS FIRST

Functional or non-functional?

Think • Discuss • Share

Functional vs. non-functional

01

FUNCTIONAL

Customers can place orders. Drivers can accept deliveries. Managers can mark dishes unavailable.

02

NON-FUNCTIONAL

95% of searches finish within 2 seconds. User accounts require secure authentication.

DISCUSS FIRST

What about business and user requirements?

Think • Discuss • Share

The same product, different levels

01

BUSINESS GOAL

Reduce phone orders and improve order accuracy within six months.

02

USER NEED

As a customer, I want to edit my cart so I can correct an order before checkout.

DISCUSS FIRST

What if a requirement comes from outside the client?

Think • Discuss • Share

Not every requirement is a feature

Some rules come from laws, industry standards, or the environment.

01

REGULATORY / LEGAL

External laws and regulations that apply to the product: sales tax, consumer protection, privacy, and labor rules.

02

INDUSTRY STANDARDS

Industry or technical rules the product must follow when applicable: healthcare data exchange, imaging formats, accessibility.

Example: Jordanian sales tax

A requirement may be imposed by law, not requested by a customer.

STANDARD GENERAL SALES TAX

16%

For many taxable goods and services in Jordan.

Some products/services have reduced rates, special rates, a zero rate, or exemptions.

Example requirement: Apply the legally applicable tax rate by item category.

Source: Jordan Income and Sales Tax Department (ISTD).

DISCUSS FIRST

What special requirements do medical systems face?

Think • Discuss • Share

Medical software has extra constraints

The exact obligations depend on the system, country, and use case.

HL7 FHIR

Exchange patient records in compatible formats.

DICOM

Store and exchange medical images.

PRIVACY RULES

Limit access to confidential patient information.

AUDIT TRAILS

Record who accessed or changed patient data.

Other categories worth knowing

These categories overlap and are often documented as constraints.

SECURITY

Bank app: require MFA for high-risk transfers.

PRIVACY

Photo app: ask permission before using location.

ACCESSIBILITY

Campus portal: keyboard navigation and readable labels.

INTEGRATION

Store: connect orders to an external payment gateway.

DATA / RETENTION

Clinic: retain records for the required period.

HARDWARE / PLATFORM

Kiosk: operate on a touchscreen without a keyboard.

DISCUSS FIRST

How would you discover these hidden constraints?

Think • Discuss • Share

Ask who sets the rules

Stakeholders are not always the people using the app.

- Which laws and regulations apply to this product?
- What standards do we need to support or comply with?
- Which third-party systems must we integrate with?
- What data must we protect, log, or retain?
- Which rules vary by country, industry, or customer?

DISCUSS FIRST

How do engineers discover requirements?

Think • Discuss • Share

Requirements elicitation

- Interviews: ask stakeholders about goals and pain points
- Observation: watch current work in context
- Surveys and workshops: gather and align perspectives
- Prototypes: expose misunderstandings early
- Workflow diagrams: map activities, decisions, and handoffs

DISCUSS FIRST

What makes a stakeholder question useful?

Think • Discuss • Share

Ask about problems, not just features

VAGUE / UNTESTABLE

“Do you want notifications?”

CLEAR / TESTABLE

“When do you need to know something has changed, and what action should follow?”

Ask: Could two engineers interpret this differently? Could a tester verify it?

DISCUSS FIRST

What is a workflow diagram?

Think • Discuss • Share

Workflow = steps + decisions + actors

- Shows how a task moves from start to completion
- Makes handoffs and decision points visible
- Reveals missing rules, data, and failure paths
- Can describe current practice (as-is) or a proposed process (to-be)

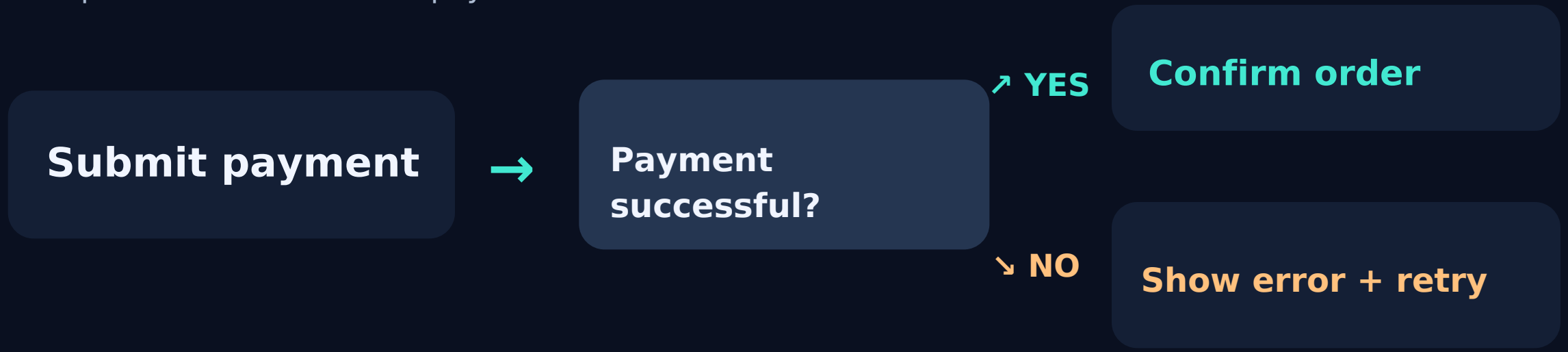
DISCUSS FIRST

How can a workflow reveal missing requirements?

Think • Discuss • Share

Look for decisions and exceptions

Example: a customer tries to pay for an order.



New requirements: payment error messages, retry behavior, duplicate-charge prevention.

DISCUSS FIRST

Can one application have different workflows?

Think • Discuss • Share

One application. Multiple stakeholders.

- Customer: wants to order and track food
- Restaurant: wants to prepare and hand over orders
- Driver: wants to pick up and deliver correctly
- Platform admin: wants to monitor and resolve issues

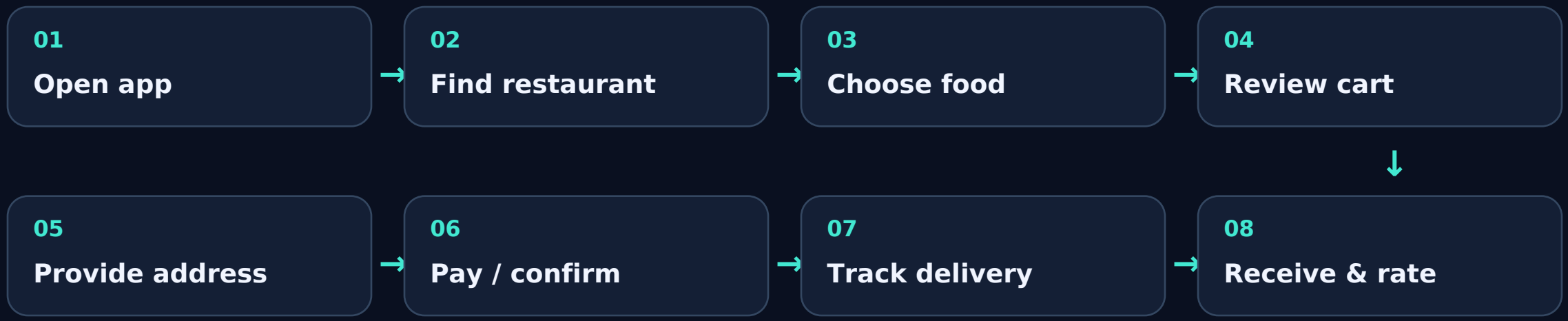
DISCUSS FIRST

What does the customer workflow look like?

Think • Discuss • Share

Customer journey

One system. A different user goal and journey.



REQUIREMENTS DISCOVERED

Cart editing • Payment choices • Tracking • Order history

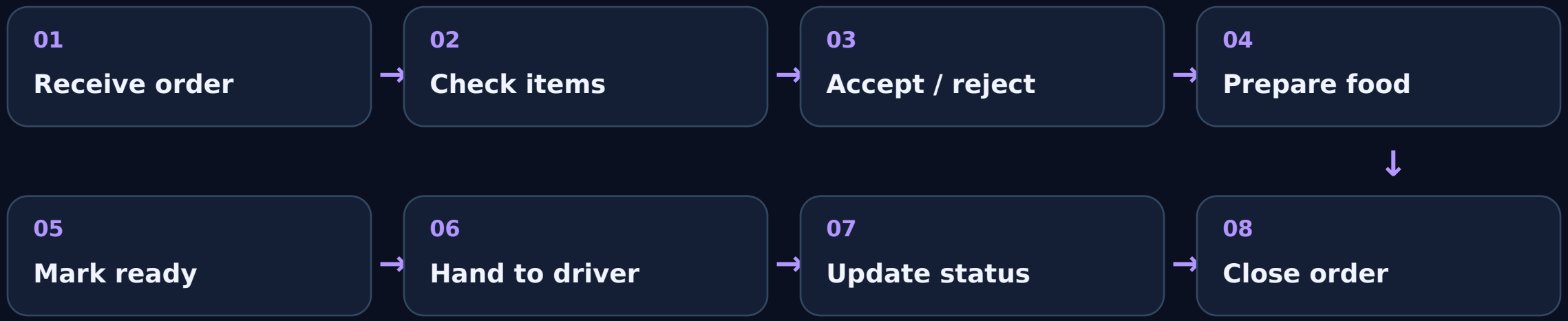
DISCUSS FIRST

What does the restaurant workflow look like?

Think • Discuss • Share

Restaurant journey

One system. A different user goal and journey.



REQUIREMENTS DISCOVERED

Order notifications • Stock control • Status changes • Rejection reasons

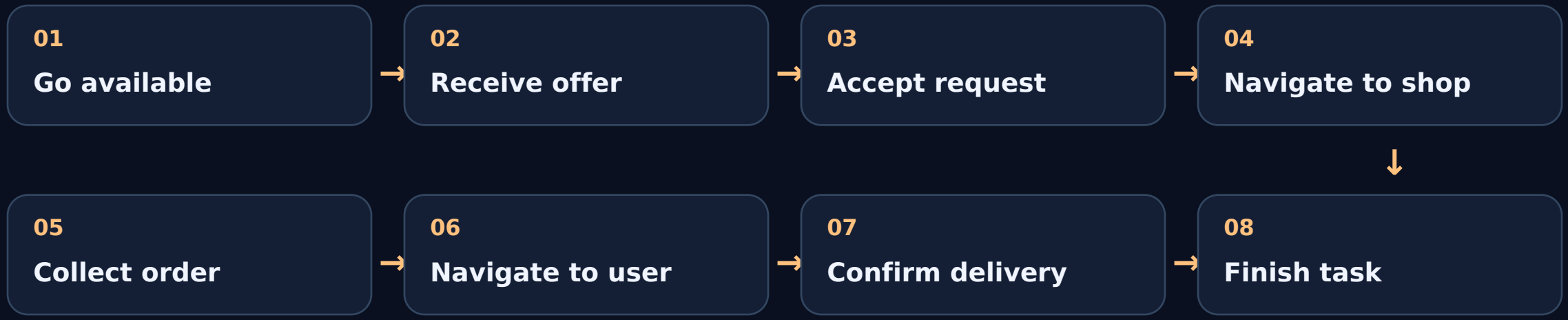
DISCUSS FIRST

What does the driver workflow look like?

Think • Discuss • Share

Driver journey

One system. A different user goal and journey.



REQUIREMENTS DISCOVERED

Availability status • Navigation • Pickup verification • Proof of delivery

DISCUSS FIRST

What happens when workflows overlap?

Think • Discuss • Share

Handoffs must agree

Customer, restaurant, and driver share a single order lifecycle.

CUSTOMER

Places order

→

RESTAURANT

Accepts & prepares

→

DRIVER

Collects & delivers

Ask: who owns each status? Who is notified? What if an order is canceled?

DISCUSS FIRST

What did our three workflows uncover?

Think • Discuss • Share

Requirements come from every role

- Different permissions, screens, and responsibilities
- Shared order statuses and coordinated handoffs
- Exceptions: rejection, delayed pickup, failed delivery
- More stakeholders = more needs to validate

DISCUSS FIRST

How do we write a good requirement?

Think • Discuss • Share

Good requirements are verifiable

- Clear and unambiguous
- Necessary and relevant
- Feasible and consistent
- Testable with defined acceptance criteria
- Traceable to a stakeholder need

Vague vs. measurable

VAGUE / UNTESTABLE

“The app should be fast.”

CLEAR / TESTABLE

“95% of menu searches return within 2 seconds under 200 concurrent users.”

Ask: Could two engineers interpret this differently? Could a tester verify it?

DISCUSS FIRST

What is a user story?

Think • Discuss • Share

Describe the value from a user’s view

As a [role],

I want [capability],

so that [benefit].

Example: As a customer, I want to reorder a past meal so I can save time.

DISCUSS FIRST

Can AI help us discover and organize requirements?

Think • Discuss • Share

Prompt engineering as a support skill

EXAMPLE PROMPT

“Act as a business analyst for a food-delivery platform. Interview me one question at a time. Identify stakeholders, workflows, functional requirements, and non-functional constraints. Flag assumptions instead of inventing answers. Then draft a requirements list with testable acceptance criteria.”

AI suggests and structures. Stakeholders validate.

DISCUSS FIRST

Class challenge: can you interview a client?

Think • Discuss • Share

Mini-workshop / 12 minutes

Work in pairs: one stakeholder, one requirements engineer.

- Client: “I need a university attendance system.”
- Engineer: ask about roles, real workflows, and exceptions
- Produce: 3 functional + 2 non-functional requirements
- Draw: one workflow including at least one decision

DISCUSS FIRST

What comes after requirements?

Think • Discuss • Share

UI / UX DESIGN

Turn validated needs into user flows, wireframes, and interfaces.

Build the right thing

before building the thing right.

UNDERSTAND → SPECIFY → VALIDATE → DESIGN